Machine Learning

Dr. Shylaja S S

Professor
Department of Computer Science & Engineering

Acknowledgement : Dr. Rajanikanth K (Academic Advisor, PESU)
Prof. Preet Kanwal (Associate Professor, PESU)
Teaching Assistant (Arya Rajiv Chaloli)

MACHINE LEARNING

⇒ Gaussian Mixture Models

Dr. Shylaja S S
Department of Computer Science and Engineering

Why GMMs?

- By modifying the mean and variance of a Gaussian, it can be made to look like any sort of ellipse (refer to the previous slide’s animation to see how the mean and variance affect the shape of a Gaussian)

- Hence using GMM allows one to build clusters in any elliptical shape (not just spherical, which is what K-Means tends to form as we shall see soon)

- Also, since GMM is a *soft clustering* algorithm, it’s more flexible than K-Means which is a hard clustering algorithm

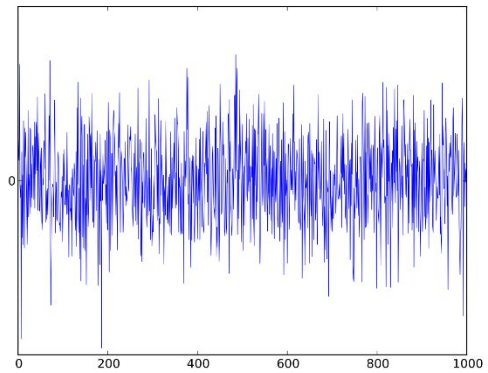
- The power of the Gaussian distribution specifically is its popularity (thanks to the *Central Limit Theorem*)

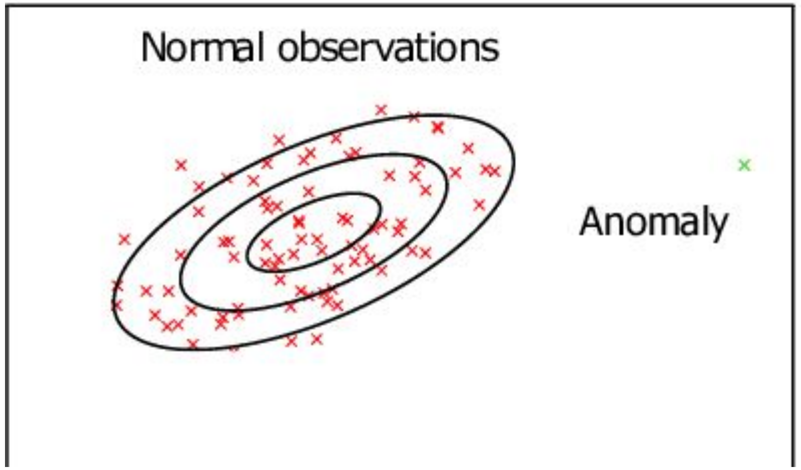
Applications of GMM

- Signal Processing: Additive White Gaussian Noise (AWGN) is a model like the GMM analyzed here, EM is used to learn this model

- Anomaly detection, a.k.a., outlier analysis

- Various classification tasks (language identification, genre identification in music)





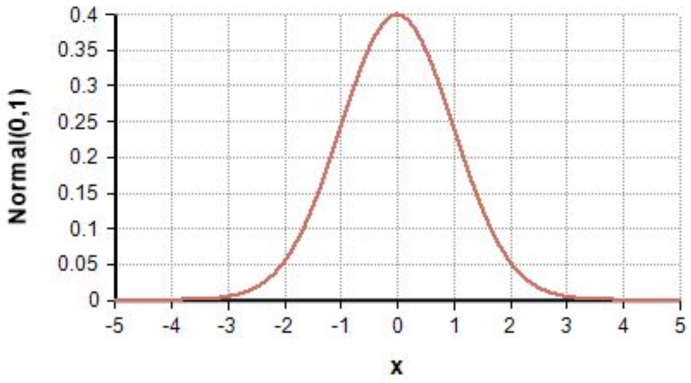
Discussion: Gaussian Distribution

The univariate normal or Gaussian distribution has the following probability density function:

$$P(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \frac{1}{E} \exp\left(-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2\right) = \frac{1}{E} \exp\left(-z^2/2\right), \quad E = \sqrt{2\pi}\sigma$$

Maximum Likelihood Estimates for mean and variance:

$$\hat{\mu} = \overline{x} \equiv \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$



Background: Multivariate Normal Distribution

Multivariate normal distributions has the following pdf:

$$P(\mathbf{x}|\mu,\Sigma) = \frac{1}{E_d}\exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)\right), \quad E_d = (2\pi)^{d/2}\sqrt{|\Sigma|}$$

- **X** = $(x_l, x_2, \ldots, x_d)^T \in R^d$; (**X** is d – dimensional feature vector)
- Parameters:
- Means vector: $\mu = (\mu_1, \mu_2, \ldots, \mu_d)^T$;
- $\sum$: d x d covariance matrix; $\sum^{-1}$ is inverse of $\sum$; $|\sum\}|$: determinant of $\sum$
- Maximum Likelihood Estimation of the Parameters μ and $\sum$: (Standard Statistics texts make use of some additional lemmas and theorems to derive the ML estimates! It is messy even then!!

Example: Multivariate Statistical Analysis by Anderson

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{X}_i \qquad \widehat{\mathbf{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^T$$

Mixture of Multivariate normal distributions:

- K multivariate Gaussian distributions; each with its own means vector μ_k and covariance matrix Σ_k ; $k = 1, \ldots, K$
- Data points are generated by these K distributions, with the proportion of points coming from these distributions governed by the priors $\pi = (\pi_1, \pi_2, \ldots, \pi_K)$.
- If each instance were labeled with the Gaussian from which it came, it would be easy to estimate the parameters of the Gaussians. From all the points labeled j, we estimate μ_j and Σ_j as already discussed.

What to do if the instance labels are not known?

This is the problem we are discussing in this session. (Chicken and Egg problem again!)

Solution:

Use EM Algorithm!

{

- We will not derive all the equations formally!
- We will use 2 Gaussians to illustrate the method and generalize it to the general case in an intuitive way.

}

GMM (2)

- The set of instances is generated by K Gaussians as follows:
  - Select one among the K Gaussians according to prior distribution $\pi = \{\pi_1, \pi_2, \ldots, \pi_K\}$
  - Generate a random data instance from the selected Gaussian.
- *Each sample is generated by exactly one Gaussian only, though we do not know which one!*
- How do we formalize this problem?
- Introduce the latent Boolean variables z_{ij}, i = 1 to n; j = 1 to k
  We associate one Boolean Vector $\mathbf{Z_i}$ = (zi1, zi2, …, zik) with the data instance $\mathbf{X_i}$.
  For each data point $\mathbf{X_i}$, exactly one of the k Boolean variables z_{ij}, j = 1 to k will be 1 and the rest will be 0. If the non-zero variable is z_{im} , then the data sample $\mathbf{X_i}$ is assumed to have been generated by Gaussian m.
- Now the GMM model can be expressed as:

$$P(\mathbf{X_i}, \mathbf{Z_i} \mid \boldsymbol{\theta}) = \sum_{j=1}^{k} z_{ij} \; \pi_j \frac{1}{(2\pi)^{d/2}\sqrt{|\boldsymbol{\Sigma}_j|}} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_j)^{\mathrm{T}} \boldsymbol{\Sigma}_j^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_j)\right)$$

θ represents all the model parameters: **π, μ,** and ∑.

Generalization – Multivariate K Gaussians

$$P(\mathbf{x}_l, \mathbf{z}_l|\theta) = \sum_{j=1}^{K} z_{lj}\tau_j \frac{1}{(2\pi)^{d/2}\sqrt{|\Sigma_j|}} \exp\left(-\frac{1}{2}(\mathbf{x}_l - \boldsymbol{\mu}_j)^{\mathrm{T}}\Sigma_j^{-1}(\mathbf{x}_l - \boldsymbol{\mu}_j)\right)$$

τ_j are the priors

Log Likelihood Function:

$$= \mathbb{E}\left[\sum_{l=1}^{n} \ln P(\mathbf{x}_l \cup \mathbf{z}_l|\theta) \middle| \mathrm{X}, \theta^t\right]$$

$$= \mathbb{E}\left[\sum_{l=1}^{n}\sum_{j=1}^{K} z_{lj} \ln\left(\tau_j \frac{1}{(2\pi)^{d/2}\sqrt{|\Sigma_j|}} \exp\left(-\frac{1}{2}(\mathbf{x}_l - \boldsymbol{\mu}_j)^{\mathrm{T}}\Sigma_j^{-1}(\mathbf{x}_l - \boldsymbol{\mu}_j)\right)\right) \middle| \mathrm{X}, \theta^t\right]$$

$$= \sum_{l=1}^{n}\sum_{j=1}^{K} \mathbb{E}\left[z_{lj}|\mathrm{X}, \theta^t\right]\left(\ln\tau_j - \frac{d}{2}\ln(2\pi) - \frac{1}{2}\ln|\Sigma_j| - \frac{1}{2}(\mathbf{x}_l - \boldsymbol{\mu}_j)^{\mathrm{T}}\Sigma_j^{-1}(\mathbf{x}_l - \boldsymbol{\mu}_j)\right)$$

E Step:

$$E[z_{ij}|\mathrm{X},\theta^t] = \frac{\tau_j^t f(\mathrm{x}_i|\mu_j^t,\Sigma_j^t)}{\sum_{k=1}^K \tau_k^t f(\mathrm{x}_i|\mu_k^t,\Sigma_k^t)}$$

M Step:

$$\tau_j^{t+1} = \frac{E_j}{\sum_{k=1}^K E_k} = \frac{1}{n}\sum_{i=1}^{n} E[z_{ij}|\mathrm{X},\theta^t]$$

M Step (continued):

$$\mu_j^{t+1} = \frac{1}{E_j}\sum_{l=1}^{n} \mathrm{E}\left[z_{lj} \middle| \mathrm{X}, \theta^t\right]\mathrm{x}_l = \frac{\sum_{l=1}^{n} \mathrm{E}\left[z_{lj} \middle| \mathrm{X}, \theta^t\right]\mathrm{x}_l}{\sum_{l=1}^{n} \mathrm{E}\left[z_{lj} \middle| \mathrm{X}, \theta^t\right]}$$

$$\Sigma_j^{t+1} = \frac{1}{E_j}\sum_{l=1}^{n} \mathrm{E}\left[z_{lj} \middle| \mathrm{X}, \theta^t\right](\mathrm{x}_l - \mu_j^{t+1})(\mathrm{x}_l - \mu_j^{t+1})^{\mathrm{T}}$$

$$= \frac{\sum_{l=1}^{n} \mathrm{E}\left[z_{lj} \middle| \mathrm{X}, \theta^t\right](\mathrm{x}_l - \mu_j^{t+1})(\mathrm{x}_l - \mu_j^{t+1})^{\mathrm{T}}}{\sum_{l=1}^{n} \mathrm{E}\left[z_{lj} \middle| \mathrm{X}, \theta^t\right]}$$

Machine Learning

Dr. Shylaja S S

Department of Computer Science & Engineering

shylaja.sharath@pes.edu